Department of Computer Science
Faculty of Computing
UNIVERSITI TEKNOLOGI MALAYSIA

SUBJECT NAME:	**COMPUTER ORGANIZATION AND ARCHITECTURE**
SUBJECT CODE:	**SECR 1033**
SEMESTER:	**2 – 2023/2024**
LAB TITLE:	**Lab 2: Arithmetic Equations & Operations**
STUDENT INFO :	Execute the lab in group of two. **Student 1**: *No. 1, 3, 5* **Student 2**: *No 2, 4, 6* **Name 1: CHOH JING YI** **Metric No: A23CS0296** **Link for Video Demo: https://youtu.be/Y450a28vFr8** **Name 2: HARINI A/P SANGARAN** **Metric No: A23CS0081** **Link for Video Demo: https://youtu.be/Y450a28vFr8**
SUBMISSION DATE & ITEMS:	**Duration of submission :-** 2 weeks **Submission items in elearning:-** 1. Lab 2 exercise sheet/file (in .pdf), with the links for demo video 5 - 10min, on the cover page. 2. The assembly programs (in .asm).

MARKS: _____

Arithmetic Equation Coding in Assembly Language

Q1. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 1 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

```
INCLUDE Irvine32.inc
.data
var1 word 1
var2 word 9

.code
main PROC
        mov ax, var1     ; LINE1
        mov bx, var2     ; LINE2
        xchg ax, bx      ; LINE3
        mov var1, ax     ; LINE4
        mov var2, bx     ; LINE5
        call DumpRegs
        exit
main ENDP
END main
```

Answer Q1

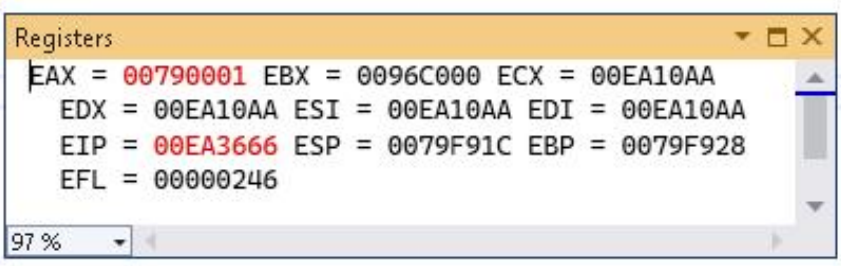
a) Fill (Write) in the contents for the related register in each line:

Table 1

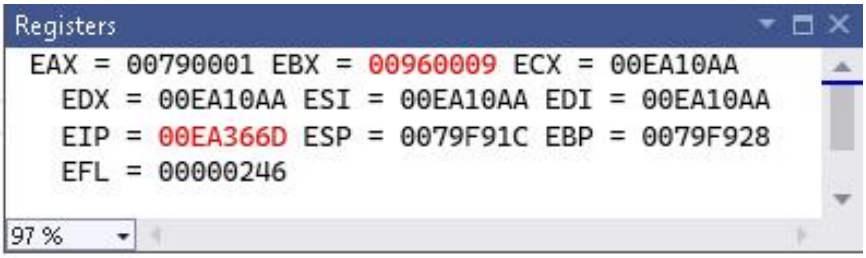
LINE1	**AX = 0001h** **var1 = 0001h**	Move the value of var1 (1d) into register AX
LINE2	**BX =** 0009h **var2 =** 0009h	Move the value of var2(9d) into register BX
LINE3	**AX =** 0009h **BX =** 0001h	Exchange the value between register AX and BX
LINE4	**AX =** 0009h **var1 =** 0009h	Move the value of register AX into variable var1
LINE5	**BX =** 0001h **var2 =** 0001h	Move the value of register BX into variable var2

b) Paste here screenshot of all registers' content after each LINE is executed:

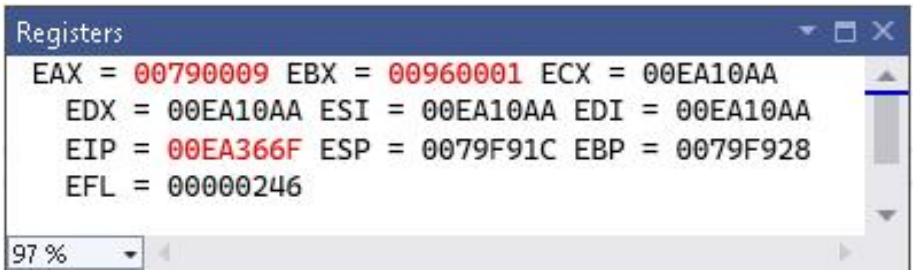
LINE1:



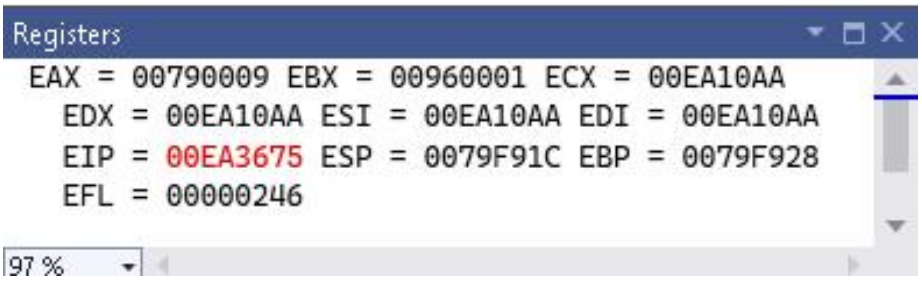
LINE2:



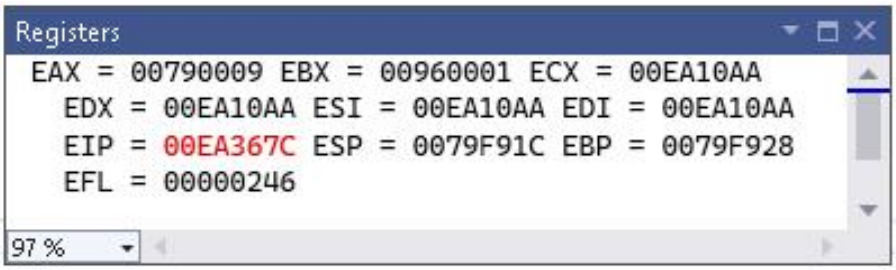
LINE3:



LINE4:



LINE5:



Q2. Execute the program below. Determine output of the program by inspecting the content of the related registers and watches.

a) Fill in Table 2 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: Rval = (-Xval + (Yval – Zval)) + 1

```
include irvine32.inc

.data
Rval DWORD ?
Xval DWORD 26
Yval DWORD 30
Zval DWORD 40

.code
main proc
      mov eax,Xval       ; LINE1
      neg eax            ; LINE2
      mov ebx,Yval       ; LINE3
      sub ebx,Zval       ; LINE4
      add eax,ebx        ; LINE5
      inc eax            ; LINE6
      mov Rval,eax       ; LINE7
      exit
main endp
end main
```

Answer Q2

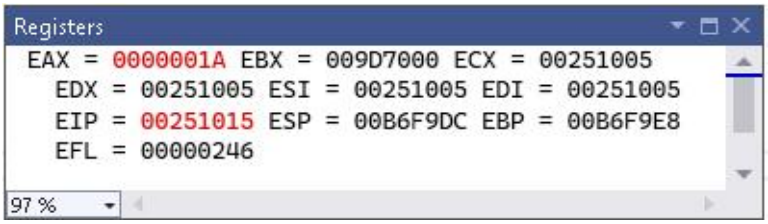
a) Fill (Write) in the contents for the related register in each line:

Table 2

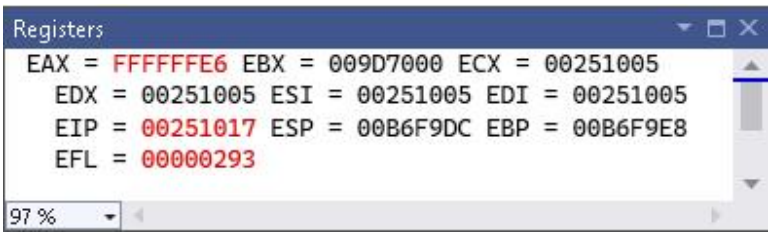
LINE1	**EAX = 0000001Ah** **Xval = 0000001Ah**	Move the value of Xval (26d) into register EAX
LINE2	**EAX =** FFFFFFE6h	Negate register EAX with 2's complement
LINE3	**EBX =** 0000001Eh **Yval =** 0000001Eh	Move the value of Yval (30d) into register EBX
LINE4	**EBX =** FFFFFF6h **Zval =** 00000028h	Move the value of Zval (30d) from register EBX
LINE5	**EAX =** FFFFFF0Ch **EBX =** FFFFFFF6h	Add value of register EBX to register EAX
LINE6	**EAX =** FFFFFFDDh	Increment of EAX by 1
LINE7	**EAX =** FFFFFFDDh **Rval =** FFFFFFDDh	Move value stored in EAX to variable Rval

b) Paste here screenshot of all registers' content after each LINE is executed:

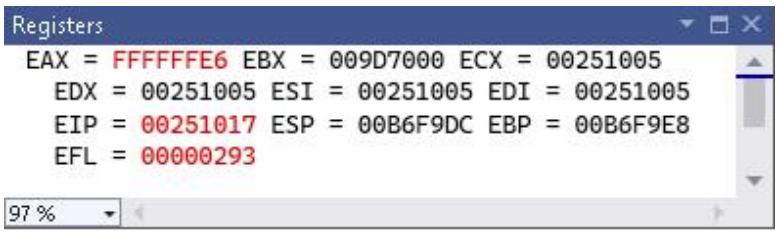
LINE1:



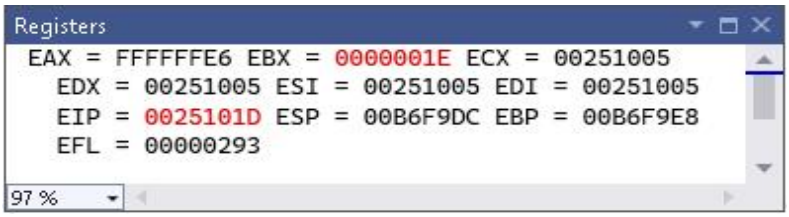
LINE2:



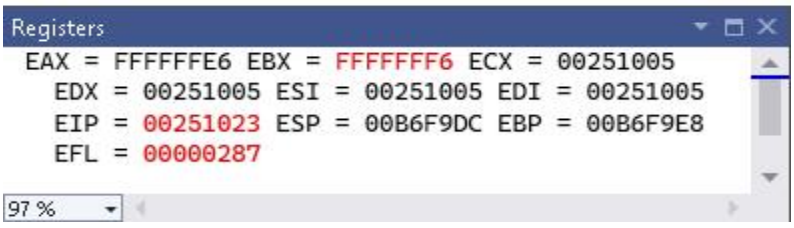
LINE3:



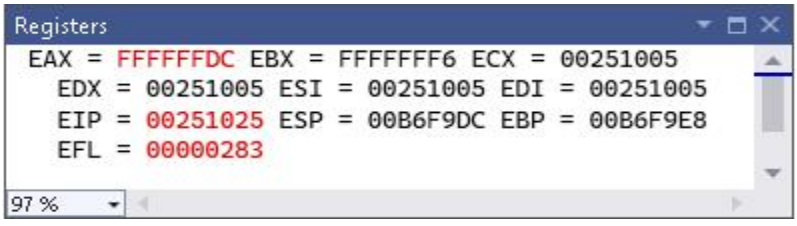
LINE4:



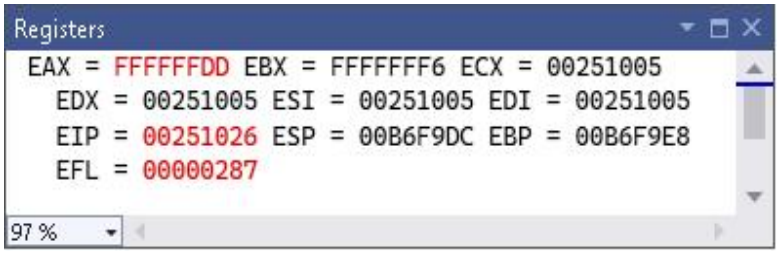
LINE5:



LINE6:



LINE7:



Q3. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 3 with the content of each register or variable on every LINE, in **Hexadecimal** (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = [(var1 * var2) + var3] - 1

```
include irvine32.inc

.data
var1 DWORD 5
var2 DWORD 10
var3 DWORD 20
var4 DWORD ?

.code
main proc
      mov eax, var1          ; LINE1
      mul var2               ; LINE2
      add eax, var3          ; LINE3
      dec eax                ; LINE4
      exit
main endp
end main
```

Answer Q3

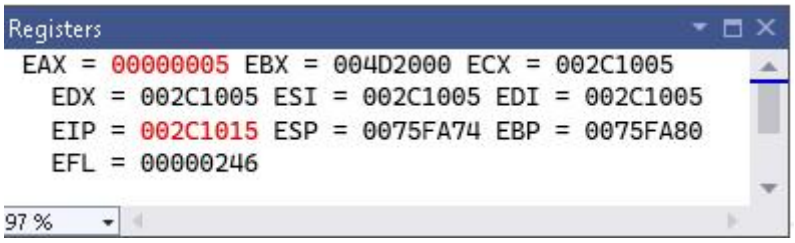
a) Fill (Write) in the contents for the related register in each line:

Table 3

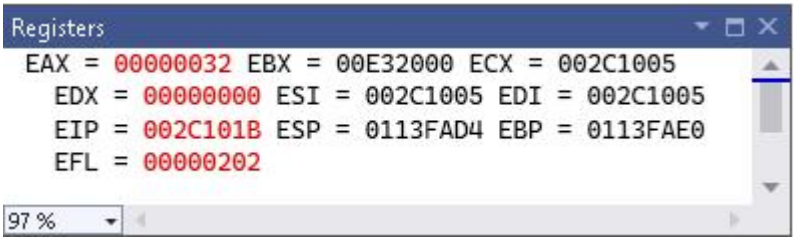
LINE1	**EAX = 00000005h** **var1 = 00000005h**	Move the value of var1 (5d) into register EAX
LINE2	**EAX =** 00000032h **var2 =** 0000000Ah	Multiply the value of var2 (10d) to register EAX
LINE3	**EAX =** 00000046h **var3 =** 00000014h	Add the value of var3 (20d) to register EAX
LINE4	**EAX =** 00000045h **var4 =** 00000000h	Decrement the value in EAX by 1

b) Paste here screenshot of all registers' content after each LINE is executed:

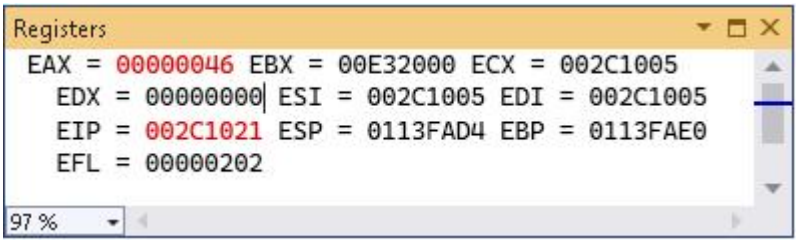
LINE1:



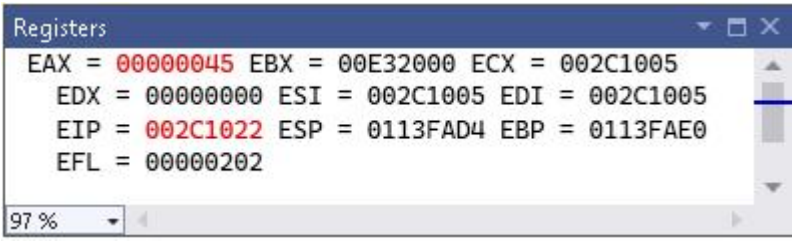
LINE2:



LINE3:



LINE4:



Q4. Execute the program below. Determine output of the program by inspecting the content of the related registers.

a) Fill in Table 4 with the content of each register or variable on every LINE, in Hexadecimal (as per the output). Please complete the comments for every LINE.

b) Paste the screenshot of all registers' content after each LINE is executed.

Arithmetic expression: var4 = (var1 * 5) / (var2 – 3)

```
include irvine32.inc
.data
      var1 WORD 40
      var2 WORD 10
      var4 WORD ?
.code
main proc
      mov ax,var1       ; LINE1
      mov bx,5          ; LINE2
      mul bx            ; LINE3
      mov bx,var2       ; LINE4
      sub bx,3          ; LINE5
      div bx            ; LINE6
      mov var4,ax       ; LINE7
      exit
main endp
end main
```

Answer Q4

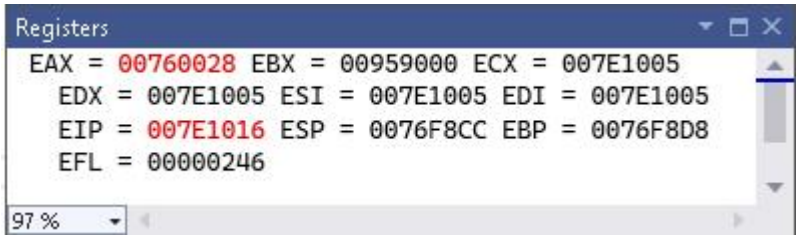
a) Fill (Write) in the contents for the related register in each line:

Table 4

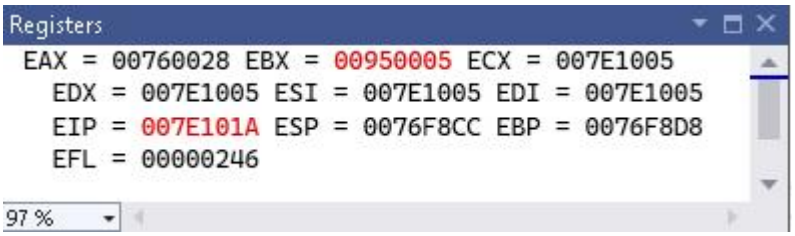
LINE1	**AX = 0028h** **var1 = 0028h**	Move the value of var1 (40d) into register AX
LINE2	**BX =** 0005h	Move the immediate value (5d) into register BX
LINE3	**AX =** 00C8h **BX =** 0005h	Multiply the value of register BX to register AX and store the value in register DX:AX
LINE4	**BX =** 000Ah **var2 =** 000Ah	Move the value of var2 (10d) into register BX
LINE5	**BX =** 0007h	Subtract the value in register BX with an immediate value (3d)
LINE6	**AX =** 001Ch **BX =** 0007h **DX =** 0004h	Divide the register AX by register BX, the quotient is then store in register AX and the remainder store in register DX
LINE7	**AX =** 001Ch **var4 =** 001Ch	Move the value of register AX into variable var4

b) Paste here screenshot of all registers' content after each LINE is executed:

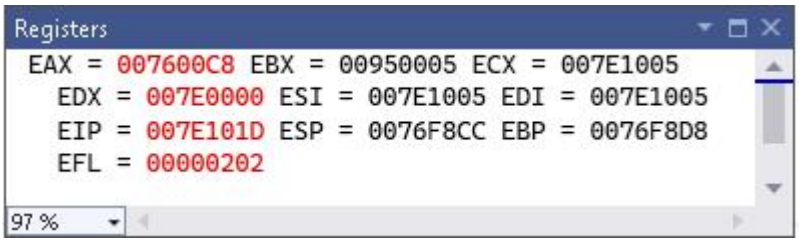
LINE1:



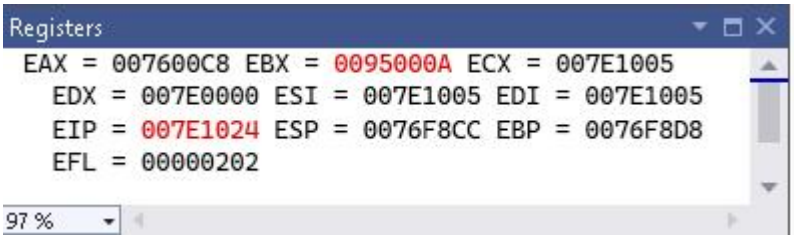
LINE2:



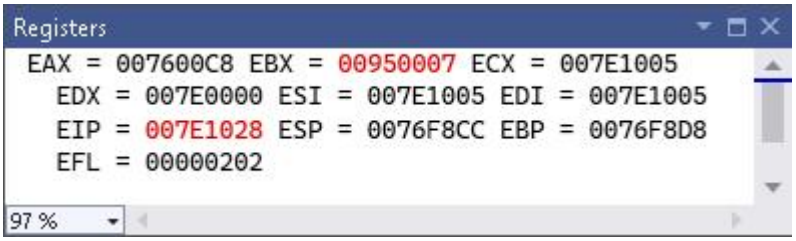
LINE3:



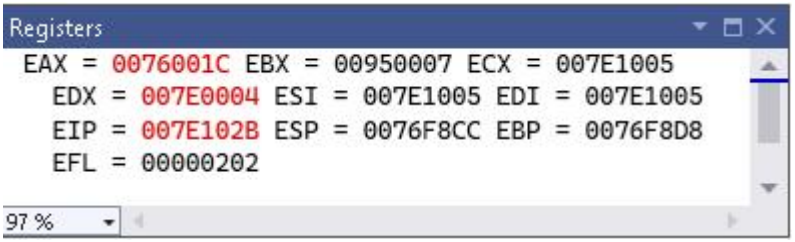
LINE4:



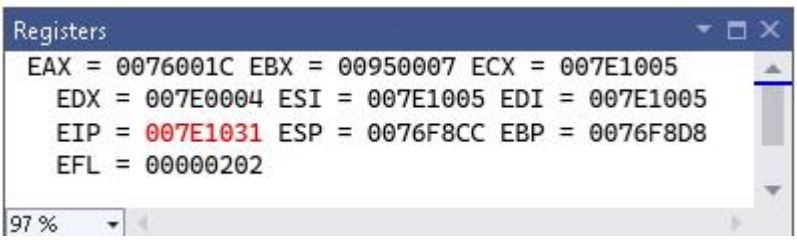
LINE5:



LINE6:



LINE7:



Short Notes for MUL CX and DIV BL:

MUL CX

a. MUL always uses AX (or its extended versions EAX or RAX) as the implicit destination register.
b. The operand size determines the size of the result:
    i. Byte-sized operand: Result in AX
    ii. Word-sized operand: Result in DX:AX
    iii. Doubleword-sized operand (32-bit mode): Result in EDX:EAX
    iv. Quadword-sized operand (64-bit mode): Result in RDX:RAX
c. The upper half of the result (DX or EDX or RDX) holds any overflow bits.
d. The Carry Flag (CF) is set if the upper half of the product is non-zero.

DIV BL

a. DIV always uses the DX:AX or EDX:EAX pair as the implicit dividend register.
b. The divisor is specified as the operand of the DIV instruction.
c. The quotient is stored in AX (for 16-bit division) or EAX (for 32-bit division).
d. The remainder is stored in DX.
e. Clear DX (or EDX for 32-bit division) before division to ensure a correct 16-bit or 32-bit dividend.
f. If the divisor is 0, a division error occurs.
g. The Overflow Flag (OF) is set if the quotient is too large to fit in the destination register.

Q5. Given the following instructions as is Code Snippet 1.

a) Write a full program to execute the Code Snippet 1.
b) What are the contents of the related registers after Code Snippet 1 is executed? Paste the screenshot of DumpReg.

; Code Snippet 1 (MUL CX)

```
MOV DX,0        ; Clear DX
MOV AX,1000h    ; Load 1000h into AX
MOV CX, 25h     ; Load 25h into CX
MUL CX          ; Multiply AX by CX, storing the result in DX:AX
```

Answer Q5

a) Screenshot of full program (.asm) :

```
1   Title lab 2
2
3   ;Authors:CHOH JING YI& HARINI A / P SANGARAN
4   ;Date:1 / 6 / 2024
5   ;Question 5
6
7   Include irvine32.inc
8   .data
9
10  .code
11  main PROC
12
13          MOV DX, 0; Clear DX
14          MOV AX, 1000h; Load 1000h into AX
15          MOV CX, 25h; Load 25h into CX
16          MUL CX; Multiply AX by CX, storing the result in DX : AX
17          call Dumpregs
18
19          exit;Exit the program
20
21  main endp
22  end main
23
```

b) Paste here the screenshot of the final registers' content (DumpReg):

```
EAX=00CF5000  EBX=00A5D000  ECX=00620025  EDX=00620002
ESI=006210AA  EDI=006210AA  EBP=00CFFCCC  ESP=00CFFCC0
EIP=00623674  EFL=00000A07  CF=1  SF=0  ZF=0  OF=1  AF=0  PF=1
```

Q6. Given the following instructions as is Code Snippet 2.

a) Write a full program to execute the Code Snippet 2.
b) What are the contents of the related registers after Code Snippet 2 is executed? Paste the screenshot of DumpReg.

```
; Code Snippet 2 (DIV BL)
MOV DX, 0       ; Clear DX to form the 16-bit dividend in DX:AX
MOV AX, 803h    ; Load the dividend (8003h) into AX
MOV BL, 10h     ; Load the divisor (10h) into BL
DIV BL          ; Divide DX:AX by BL, whereby AX=quotient & DX=remainder
```

Answer Q6

a) Screenshot of full program (.asm) :

```
1  Title lab 2
2
3  ;Authors:CHOH JING YI& HARINI A / P SANGARAN
4  ;Date:1 / 6 / 2024
5  ;Question 6
6
7  include irvine32.inc
8  .data
9
10 .code
11 main PROC
12
13         MOV DX, 0; Clear DX to form the 16 - bit dividend in DX : AX
14         MOV AX, 803h; Load the dividend(8003h) into AX
15         MOV BL, 10h; Load the divisor(10h) into BL
16         DIV BL; Divide DX : AX by BL, whereby AX = quotient & DX = remainder
17         call Dumpregs
18
19         exit;Exit the program
20
21 main endp
22 end main
23
```

b) Paste here the screenshot of the final registers' content (DumpReg):

```
EAX=01130380  EBX=00F1C010  ECX=00CB10AA  EDX=00CB0000
ESI=00CB10AA  EDI=00CB10AA  EBP=0113FD30  ESP=0113FD24
EIP=00CB3671  EFL=00000246  CF=0  SF=0  ZF=1  OF=0  AF=0  PF=1
```